

**Market Risk Measurement and Hedging
Strategy Analysis: A \$1 Billion Hong Kong
Equity Index Portfolio**

Table of Contents

A.	Investment Portfolio Description.....	3
1.	Hang Seng Index (HSI).....	3
2.	FTSE China 50 Index (XIN9).....	4
3.	Hang Seng Tech Index (HSTECH).....	5
4.	Hang Seng China Enterprises Index (HSCEI)	6
5.	Hang Seng CCI (HSCCI).....	6
B.	Risk Measurement and Analysis.....	8
1.	Summary Statistics.....	8
2.	Single Linear Regression	8
3.	VaR - Historical Simulation.....	9
4.	VaR - Delta-Normal.....	10
5.	VaR - Monte-Carlo	11
6.	Portfolio Market Risk Analysis.....	11
C.	Risk Management Strategies.....	15
1.	Keeping the Portfolio Unhedged.....	15
2.	Hedging Using E-mini S&P500 Index Futures Expiring in Sep 2025	15
3.	Hedging Using E-mini S&P500 Index Options Expiring in Sep 2025.....	17
4.	Hedging Outcome Comparison.....	17
5.	Managing Market Risk in Silo.....	18
6.	Managing Firmwide Risk.....	20
D.	Reference	23
E.	Appendix.....	25

A. Investment Portfolio Description

There are five Hong Kong stock indices chosen in our analysis: Hang Seng Index, FTSE China 50 Index, Hang Seng Tech Index, Hang Seng CEI, and Hang Seng CCI. Each index will consist of 20% of the \$1 billion portfolio.

1. Hang Seng Index (HSI)

The HSI is the most widely known Hong Kong stock market index. The index comprises the 83 most extensive and liquid stocks listed in Hong Kong. The index weights allocation across industries is as shown in Figure A.1.1.

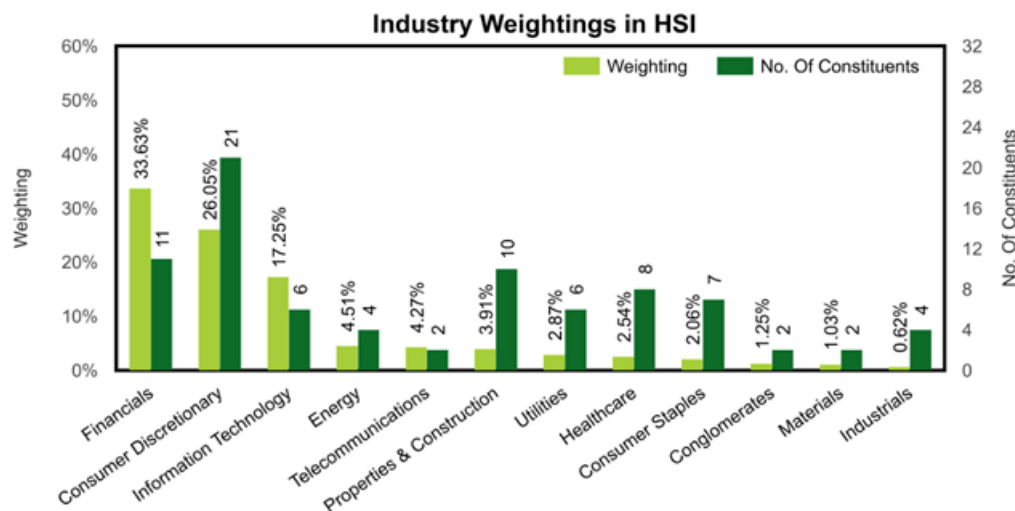


Figure A.1.1: HSI industry weightings. Source: Hang Seng Bank (2025)

The US dollar-denominated HSI return has a beta of 0.424195 when measured against the S&P500 index return. Based on the January 2, 2024, dollar price of \$2,148.11, a unit holding of 93,105.1222 will make up \$200 million of the portfolio. The index prices in Hong Kong dollars and US dollars from January 2024 to April 2025, excluding market holidays, are shown in Figure A.1.2.



Figure A.1.2: HSI denominated in HKD and USD.

2. FTSE China 50 Index (XIN9)

XIN9 is the index capturing the 50 largest and most liquid Hong Kong-listed Chinese stocks. The index weights allocation breakdown is shown as follows in Figure A.2.1:

ICB Supersector Breakdown

ICB Code	ICB Supersector	No. of Cons	Net MCap (HKDm)	Wgt %
1010	Technology	4	2,541,610	20.03
1510	Telecommunications	3	1,066,066	8.40
2010	Health Care	2	237,887	1.87
3010	Banks	10	2,708,241	21.34
3020	Financial Services	4	132,485	1.04
3030	Insurance	5	1,006,740	7.93
3510	Real Estate	1	119,576	0.94
4010	Automobiles and Parts	3	880,813	6.94
4020	Consumer Products and Services	4	891,084	7.02
4040	Retailers	2	1,437,745	11.33
4050	Travel and Leisure	1	483,638	3.81
4510	Food Beverage and Tobacco	1	113,696	0.90
5010	Construction and Materials	1	24,287	0.19
5020	Industrial Goods and Services	3	182,272	1.44
5510	Basic Resources	2	219,768	1.73
6010	Energy	3	595,726	4.69
6510	Utilities	1	48,687	0.38
Totals		50	12,690,321	100.00

Figure A.2.1: XIN9 industry weight allocation breakdown. Source: FTSE Russell (2025)

The US dollar-denominated XIN9 return has a beta of 0.369494 when measured against the S&P500 index return. Based on the January 2, 2024, dollar price of \$1,380.57, a unit holding of 144,867.7645 will make up \$200 million of the portfolio. The index price movements are shown in Figure A.2.2.

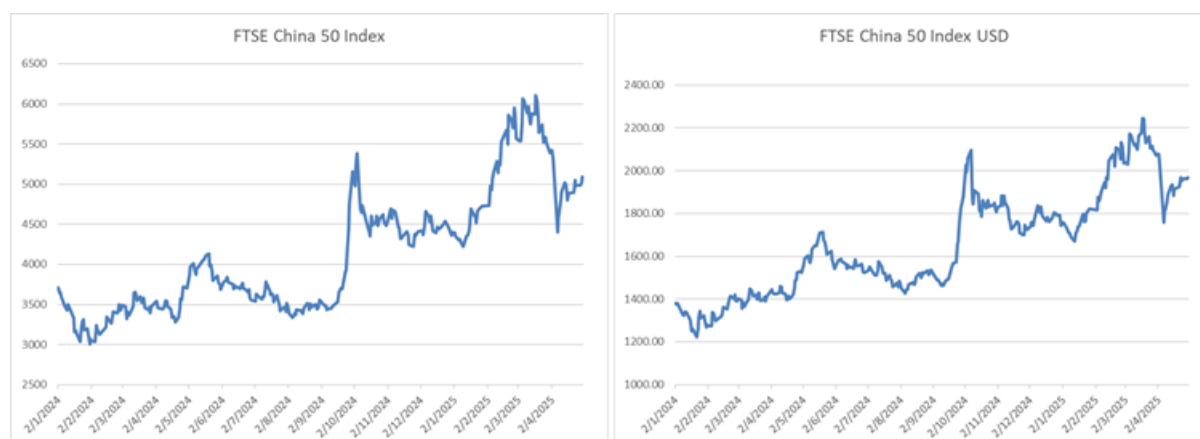


Figure A.2.2: XIN9 denominated in HKD and USD.

3. Hang Seng Tech Index (HSTECH)

HSTECH comprises the 30 largest technology companies listed on the Hong Kong Stock Exchange. Its industry weightings are displayed in Figure A.3.1.

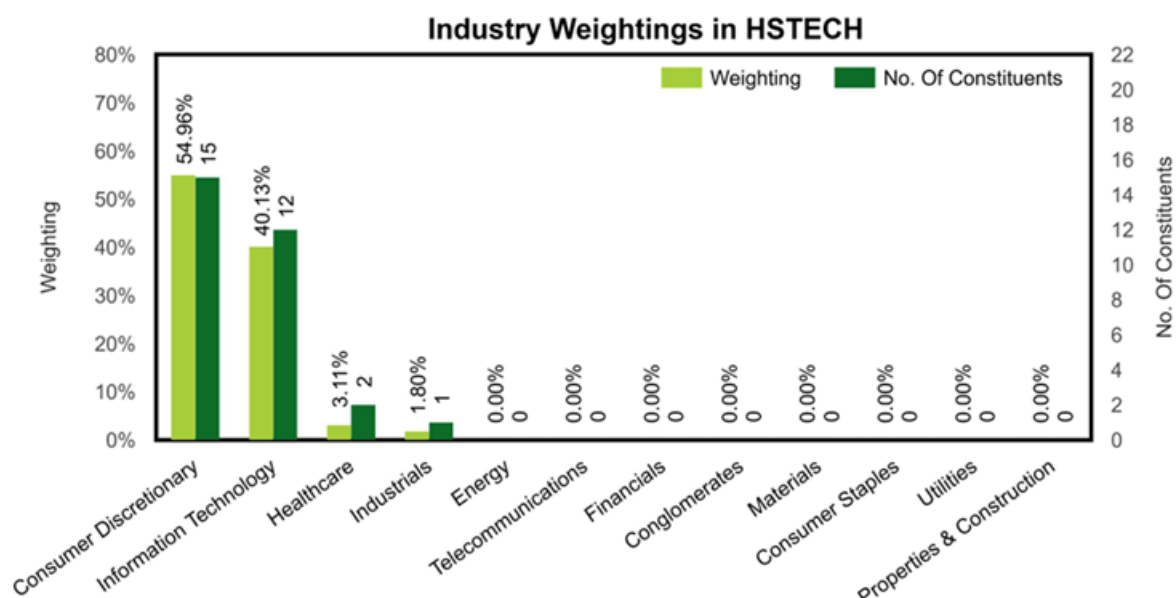


Figure A.3.1: HSTECH industry weightings. Source: Hang Seng Indexes (2025a)

The US dollar-denominated HSTECH return has a beta of 0.619348 when measured against the S&P500 index return. Based on the January 2, 2024, dollar price of \$475.31, a unit holding of 420,779.7522 will make up \$200 million of the portfolio. The index price movements are shown in Figure A.3.2.

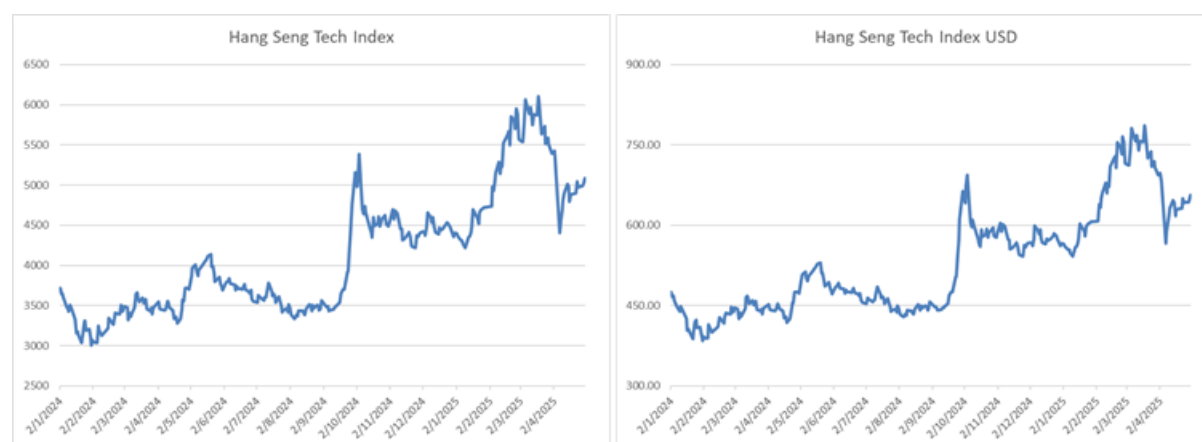


Figure A.3.2: HSTECH denominated in HKD and USD.

4. Hang Seng China Enterprises Index (HSCEI)

HSCEI is an index that reflects the performance of Hong Kong-listed Mainland China securities, comprising a total of 50 constituents. Its industry weightings are as in Figure A.4.1:

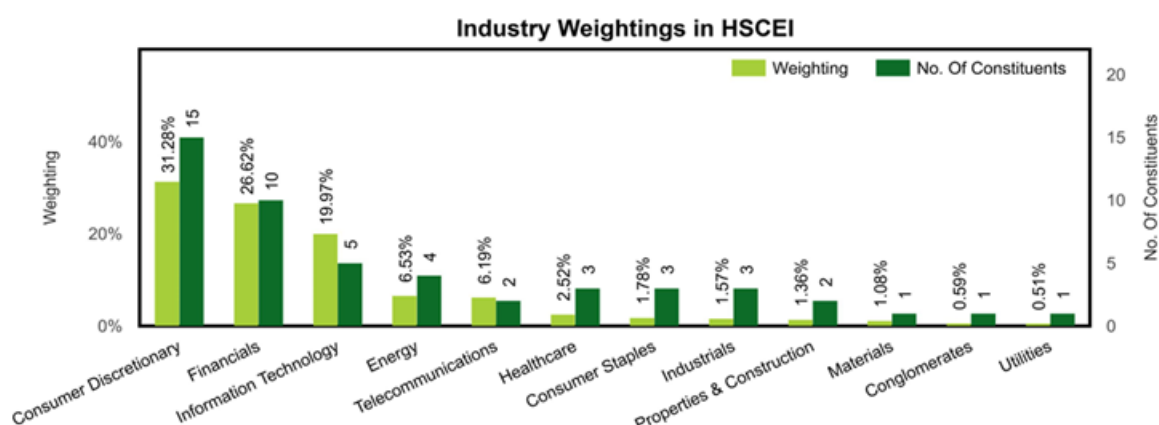


Figure A.4.1: HSCEI industry weightings. Source: Hang Seng Indexes (2025b)

The US dollar-denominated HSCEI return has a beta of 0.474400 when measured against the S&P500 index return. Based on the January 2, 2024, dollar price of \$725.81, a unit holding of 275,556.0570 will make up \$200 million of the portfolio. The index price movements are shown in Figure A.4.2.

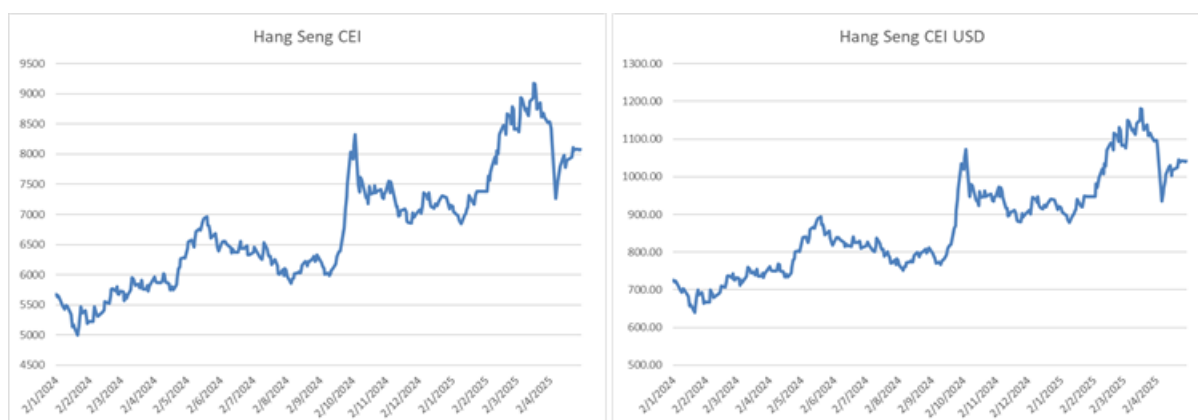


Figure A.4.2: HSCEI in HKD and USD.

5. Hang Seng CCI (HSCCI)

HSCCI provides a benchmark for Mainland-controlled companies operating outside Mainland China and listed on the Hong Kong Stock Exchange. The index comprises 25 constituents. The weightage allocations are demonstrated in Figure A.5.1.

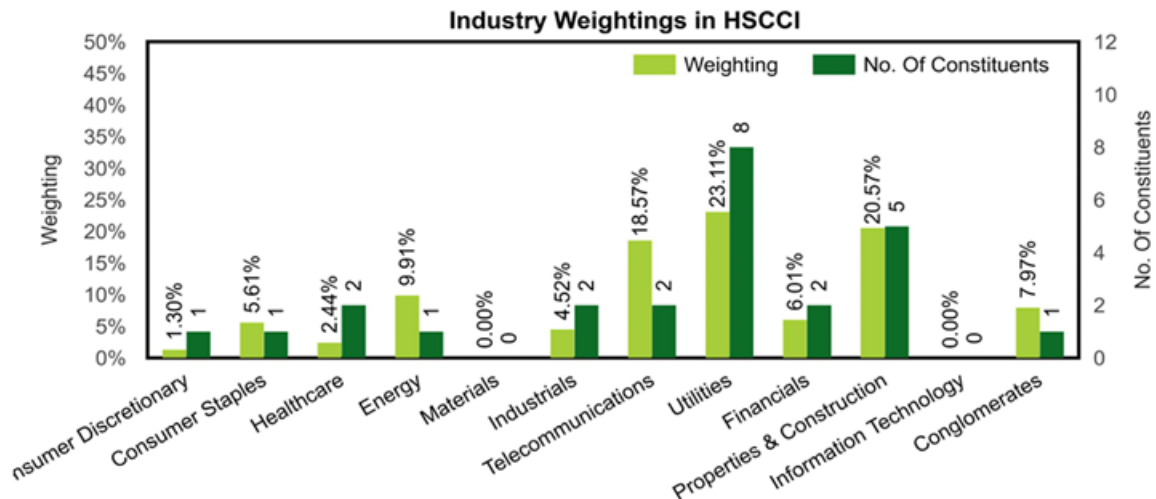


Figure A.5.1: HSCCI industry weightings. Source: Hang Seng Indexes (2025c)

The US dollar-denominated HSCCI return has a beta of 0.312630 when measured against the S&P500 index return. Based on the January 2, 2024, dollar price of \$420.53, a unit holding of 475,584.8589 will make up \$200 million of the portfolio. The index price movements are shown in Figure A.5.2.



Figure A.5.2: HSCCI in HKD and USD.

B. Risk Measurement and Analysis

1. Summary Statistics

According to Table B.1, during the observation period, we have a total of 316 market days. The minimum portfolio value is \$863,500,917.20, while the maximum is \$1,062,380,728. The mean portfolio value is \$1,001,134,987, and the median is \$1,000,079,449. The standard deviation is \$18,247,942.69. The portfolio value is negatively skewed with a skewness of -1.2974, and the kurtosis is 10.8772, suggesting a leptokurtic distribution.

Portfolio Descriptive Statistics	
Count	316
Min	863,500,917.2
Max	1,062,380,728
Mean	1,001,134,987
Median	1,000,079,449
Standard Deviation	18,247,942.69
Skewness	-1.2974
Kurtosis	10.8772

Table B.1: Portfolio descriptive statistics summary.

2. Single Linear Regression

The regression summary output is shown in Table B.2, and the regression equation is $y = 0.00018 + 0.25544x$. The R-squared value is very low at 0.013125, indicating limited explanatory power of the S&P500 HPR on the portfolio HPR. The intercept is statistically insignificant (p-value = 0.8962), suggesting that the portfolio is not outperforming S&P500 consistently after adjusting for beta. The Beta is 0.25544 and significant at 5% level (p-value = 0.0422), which implies low sensitivity towards the S&P500 HPR.

Summary Output	
R ²	0.013125
Adjusted R ²	0.009972
Observations	315

	Coefficients	Standard Error	t-stat	p-value
Intercept	0.0001844	0.0014116	0.130627	0.896154
S&P500 HPR	0.2554399	0.1251982	2.040284	0.042161

Table B.2: Summary output of single linear regression.

3. VaR - Historical Simulation

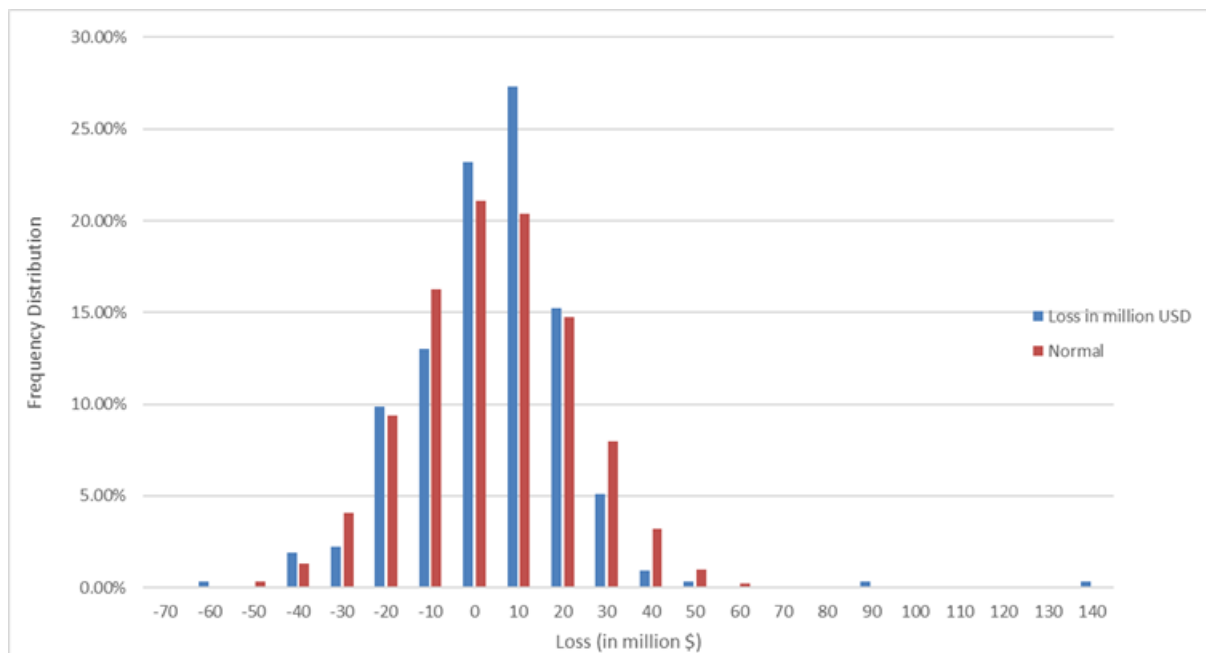
Summarising the past portfolio performance, the frequency distribution of the daily returns is as follows.

Loss in million \$	Frequency	Relative Frequency	Cumulative Frequency
-70	0	0.00%	0.00%
-60	1	0.32%	0.32%
-50	0	0.00%	0.32%
-40	6	1.90%	2.22%
-30	7	2.22%	4.44%
-20	31	9.84%	14.29%
-10	41	13.02%	27.30%
0	73	23.17%	50.48%
10	86	27.30%	77.78%
20	48	15.24%	93.02%
30	16	5.08%	98.10%
40	3	0.95%	99.05%
50	1	0.32%	99.37%
60	0	0.00%	99.37%
70	0	0.00%	99.37%
80	0	0.00%	99.37%
90	1	0.32%	99.68%
100	0	0.00%	99.68%

110	0	0.00%	99.68%
120	0	0.00%	99.68%
130	0	0.00%	99.68%
140	1	0.32%	100.00%

Table B.3.1: Daily returns frequency distribution.

Based on the relative frequency distribution in Table B.3.1, the most severe 1% loss happened approximately at the \$40 million loss bin with a cumulative frequency of 99.05% at this loss level. To be more specific, the exact $\text{VaR}_{99\%}$ loss is \$39.88 million, as determined using the Excel percentile function. The expected shortfall is \$85.97 million. The histogram of historical and normally distributed loss is visualised in Graph B.3.2.



Graph B.3.2: Histogram of loss distribution.

4. VaR - Delta-Normal

Under the Delta-normal approach, the VaR and ES are calculated using the formulas as follows:

$$\text{VaR} = \mu + \sigma Y_X \quad \text{ES} = \mu + \sigma \frac{e^{-Y_X^2/2}}{\sqrt{2\pi}(1-X)}$$

As determined earlier, the mean loss is -\$1.1386 million, and the standard deviation is \$18.2769 million. At a 99% confidence level, $Y_X = 2.326$. The result is as follows:

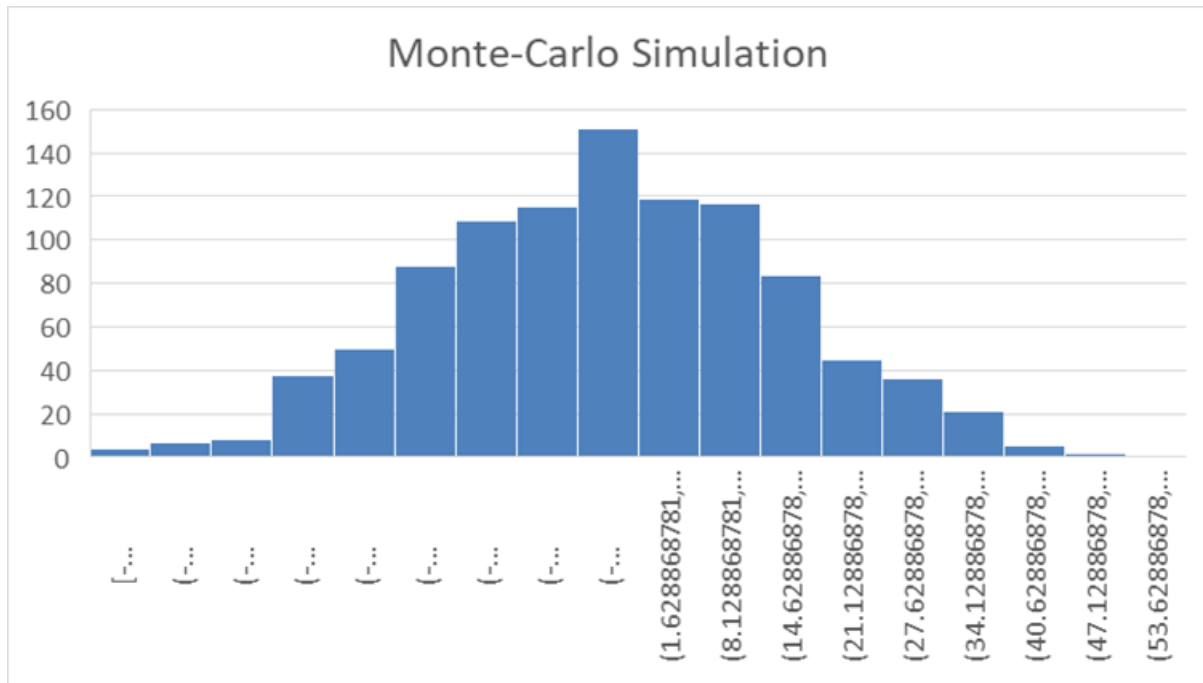
$$\text{VaR} = -1.1386 + 18.2769(2.326) = 41.3735$$

$$\text{ES} = -1.1386 + 18.2769 \frac{e^{-2.326^2/2}}{\sqrt{2\pi}(1 - 0.99)}$$

The computation shows that the $\text{VaR}_{99\%}$ is at \$41.3735 million and the ES is \$47.6127 million. As compared to the historical simulation method, the $\text{VaR}_{99\%}$ in this case is higher, but the expected shortfall is significantly lower.

5. VaR - Monte-Carlo

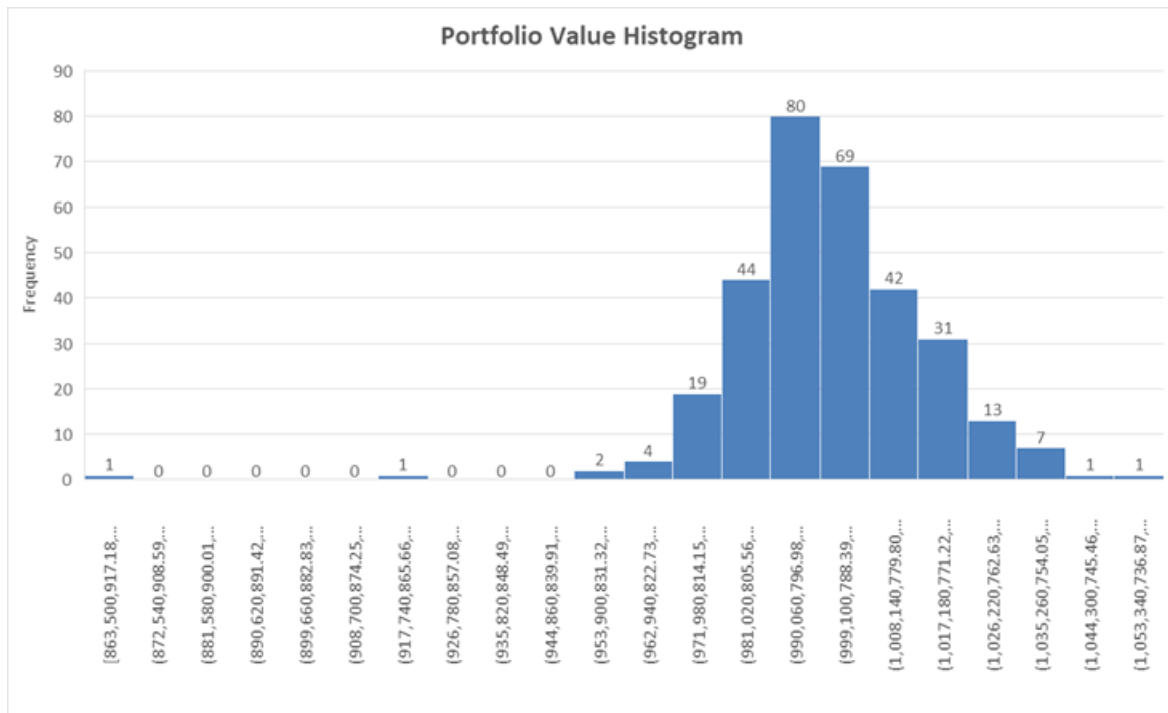
Using the Monte-Carlo method with 1000 replications, the $\text{VaR}_{99\%}$ is \$39.93 million and the ES is \$45.742 million (Appendix B.5). The portfolio losses distribution is as shown in Graph B.5.1.



Graph B.5.1: Monte-Carlo simulation histogram.

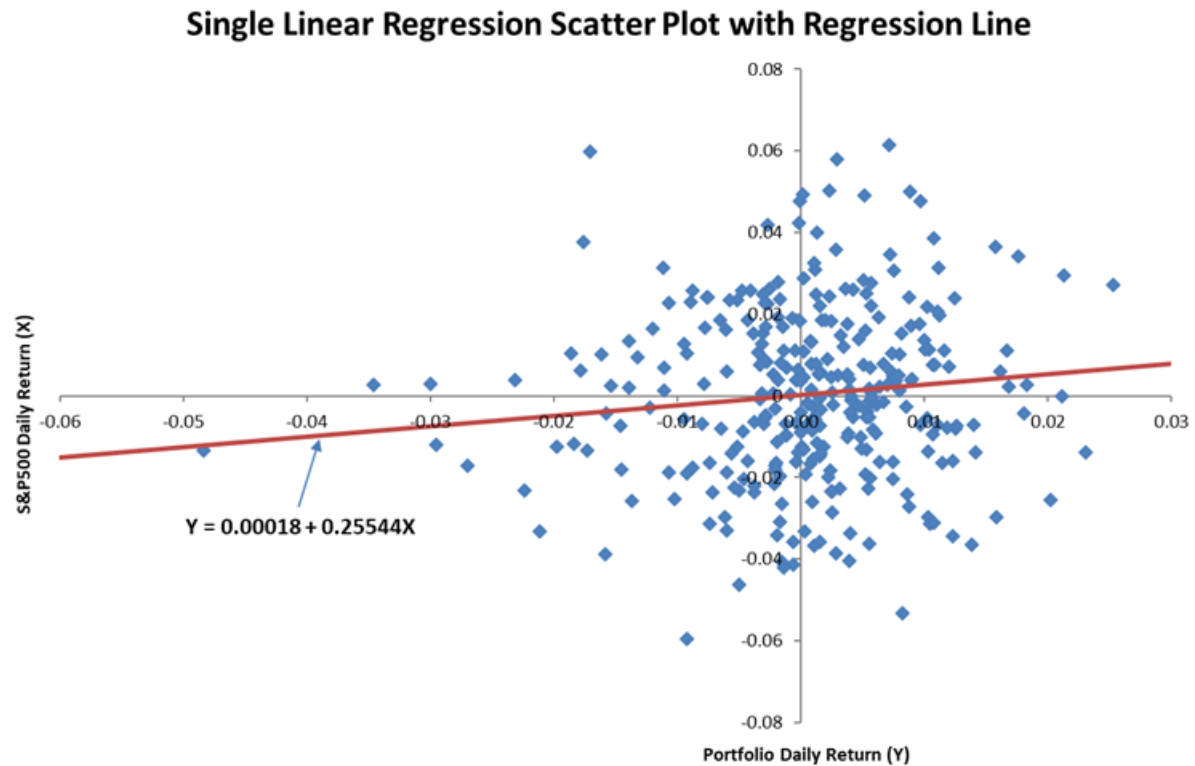
6. Portfolio Market Risk Analysis

Graph B.6.1 shows that the portfolio value during the observation period is negatively skewed and exhibits a fat tail (kurtosis = 10.8772). This suggests that a downside risk for the portfolio value is more severe and has a higher occurrence than predicted in the normal distribution. Therefore, the portfolio is subject to significant market risk.



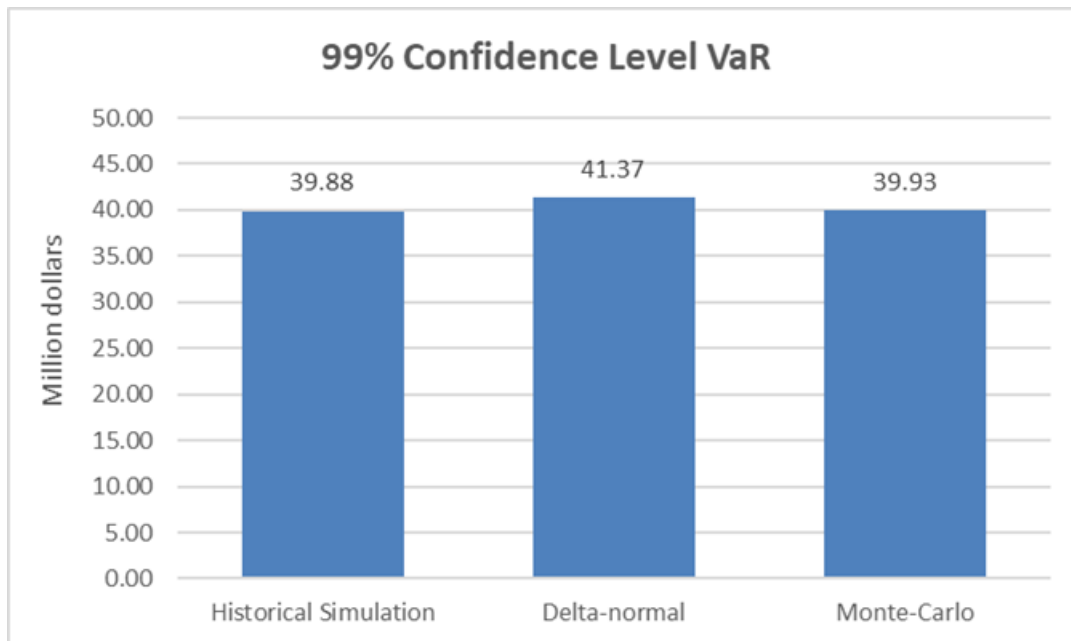
Graph B.6.1: Portfolio value histogram.

The single regression result computes that the portfolio beta is 0.25533 against the S&P500 return, suggesting a lower sensitivity to market risk. Precisely, for every 1% decrease in the S&P500 return, the portfolio return will decrease by 0.26%. This argues a defensive characteristic of the portfolio. Despite the beta being statistically significant, the regression's explanatory power is very low ($R^2 = 0.0131$), as it is only able to explain 1.31% variance in the portfolio return. As shown in Graph B.6.2, most of the data points lie far from the regression line, indicating an unreliable predictive model in measuring the portfolio's market risk exposure.



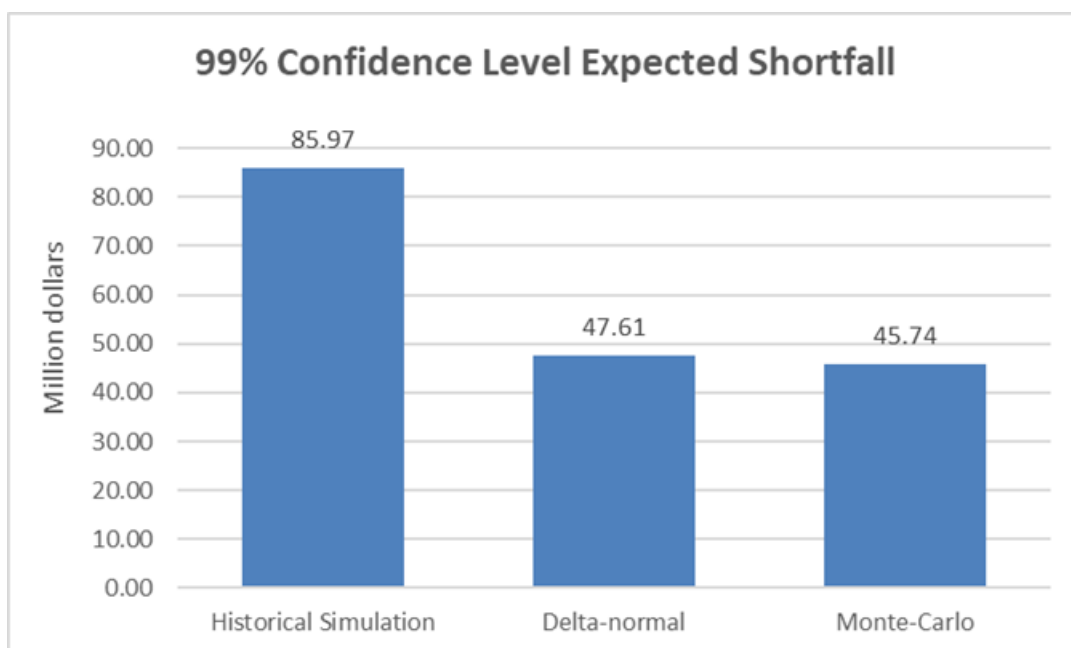
Graph B.6.2: Single linear regression scatter plot with regression line.

As we applied three methods to estimate the portfolio VaR, we obtained varying results as summarised in Graph B.6.3. The delta-normal approach produces the highest $\text{VaR}_{99\%}$ at \$41.37 million. The second approach is the Monte-Carlo approach, which determines a $\text{VaR}_{99\%}$ at \$39.93 million, followed by historical simulation with a value of \$39.88 million. The delta-normal approach is highest because it assumes normally distributed VaR. This tends to overestimate tail risk, especially when the data is negatively skewed and leptokurtic, as in our case. The Monte-Carlo method captures the non-normal characteristics of the dataset, including skewness and kurtosis. Therefore, it is more accurate than the delta-normal approach. Historical simulation has the lowest estimate but the highest accuracy for the observed data. However, the method misses tail events and is subject to window selection bias. Nevertheless, although the VaR varies using different approaches, the variation is minimal, ranging from \$39.88 to \$41.37 million. Therefore, the one-day portfolio market risk exposure at a 99% confidence level is 3.99% of the portfolio value in the best-case scenario and 4.14% in the worst-case scenario.



Graph B.6.3: $\text{VaR}_{99\%}$ using different methods.

The $\text{ES}_{99\%}$ shows significantly divergent results, as shown in Graph B.6.4. The historical simulation method generates an ES of \$85.97 million, due to some extreme shortfall cases during the observed period. This substantially contributes to the extraordinarily high ES. On the other hand, the delta-normal approach produces an ES of \$47.61 million. This is because a normal distribution assumption can underestimate the severity of historical cases. Additionally, the Monte-Carlo approach's ES is \$45.74 million, as this method smoothes out the extreme cases in reality. In summary, the ES when the losses are in the 99-percentile is \$85.97 million in the worst-case scenario and \$45.74 million in the best-case scenario.



Graph B.6.4: $\text{ES}_{99\%}$ using different methods.

C. Risk Management Strategies

In this section, we will formulate the portfolio hedging strategies, assuming the portfolio is invested as of 30 April 2025 with a value of \$1 billion, and the hedging period extends until September 2025.

1. Keeping the Portfolio Unhedged

If the fund is unhedged, it will be fully exposed to market risk, and the portfolio's value will fluctuate in response to changes in the global equity market. Since the portfolio return has a beta of 0.25544 relative to the S&P 500 return, its price movements are significantly less sensitive to fluctuations in the U.S equity market.

This strategy may benefit from low beta protection because losses in a market downturn may be relatively limited. Besides, it eliminates the need to manage hedging instruments or incur transaction costs, resulting in simplicity. However, no hedging implementation can completely shield the fund from losses in sharp declines.

S&P500 change	S&P500 Index	Portfolio Value Change	Net Portfolio Value
20%	6,682.87	+51,088,000	1,051,088,000
5%	5,847.51	+12,772,000	1,012,772,000
1%	5,624.75	+2,554,400	1,002,554,400
0%	5,569.06	0	1,000,000,000
-1%	5,513.37	-2,554,400	997,445,600
-5%	5,290.61	-12,772,000	987,228,000
-20%	4,455.25	-51,088,000	948,912,000

The maximum downside loss in a 20% market drop is approximately \$ 51.1 million USD, which is significantly lower than it would be if the beta were 1. Similarly, in a 20% increase scenario, the upside is only 51.1 million USD, meaning the portfolio trades for a lower return for lower risk. This strategy is passive and straightforward, but it may underperform in rising markets due to its low beta.

2. Hedging Using E-mini S&P500 Index Futures Expiring in Sep 2025

E-mini S&P500 index futures are standardised futures contracts that follow the trend of the S&P500 index, which is a widely used benchmark of the U.S. equity market. E-Mini S&P500 index futures are widely used to hedge against portfolio losses arising from market crashes through the sale of contracts. It is a common practice for institutions to hedge their portfolio value and prevent exposure to market risk.

With the portfolio beta of 0.25544, the number of futures needed to hedge can be calculated by

the following formula:

$$N * \beta \frac{V_A}{V_F}$$

Where

N^* = the number of futures/options required to hedge

β = the beta of the portfolio

V_A = the portfolio value invested

V_F = the value per future/option

Given that the portfolio value is \$1 billion, the S&P500 index is 5,569.06 on 30 April 2025, and the portfolio beta is 0.25544. The number of future expiring in September 2025 required to short for hedging is:

$$0.25544 \times \frac{1,000,000,000}{50 \times 5,569.06} = 917.35$$

$$\approx 918$$

Hedging Outcome:

S&P500 Change	S&P500 Index	Futures Gain/Loss	Unhedged Portfolio Gain/Loss	Net Gain/Loss	Hedged Portfolio Value
20%	6,682.87	-51,123,971	+51,088,000	-35,971	999,964,029
5%	5,847.51	-12,780,855	+12,772,000	-8,855	999,991,145
1%	5,624.75	-2,556,171	+2,554,400	-1,771	999,998,229
0%	5,569.06	0	0	0	1,000,000,000
-1	5,513.37	+2,556,171	-2,554,400	+1,771	1,000,001,771
-5%	5,290.61	+12,789,855	-12,772,000	+8,855	1,000,008,855
-20%	4,455.25	+51,123,879	-51,088,000	+35,879	1,000,035,879

3. Hedging Using E-mini S&P500 Index Options Expiring in Sep 2025

Another way to hedge a portfolio is by using an E-mini S&P 500 index option. Contrastly, as opposed to futures contracts, which require the holder to purchase or sell the underlying asset at a specified price at a specified date in the future, options offer the holder rights but not obligations to purchase or sell the underlying asset at a predetermined price on or before a predetermined date in the future. It allows for a hedge against downside risk while retaining the potential for gains.

We assume the portfolio value is \$1 billion and S&P500 index is at 5,569.06 as of 30 April 2025. The number of options expiring in September 2025 can be calculated as:

$$0.25544 \times \frac{1,000,000,000}{50 \times 5,569.06} = 917.35$$

$$\approx 918$$

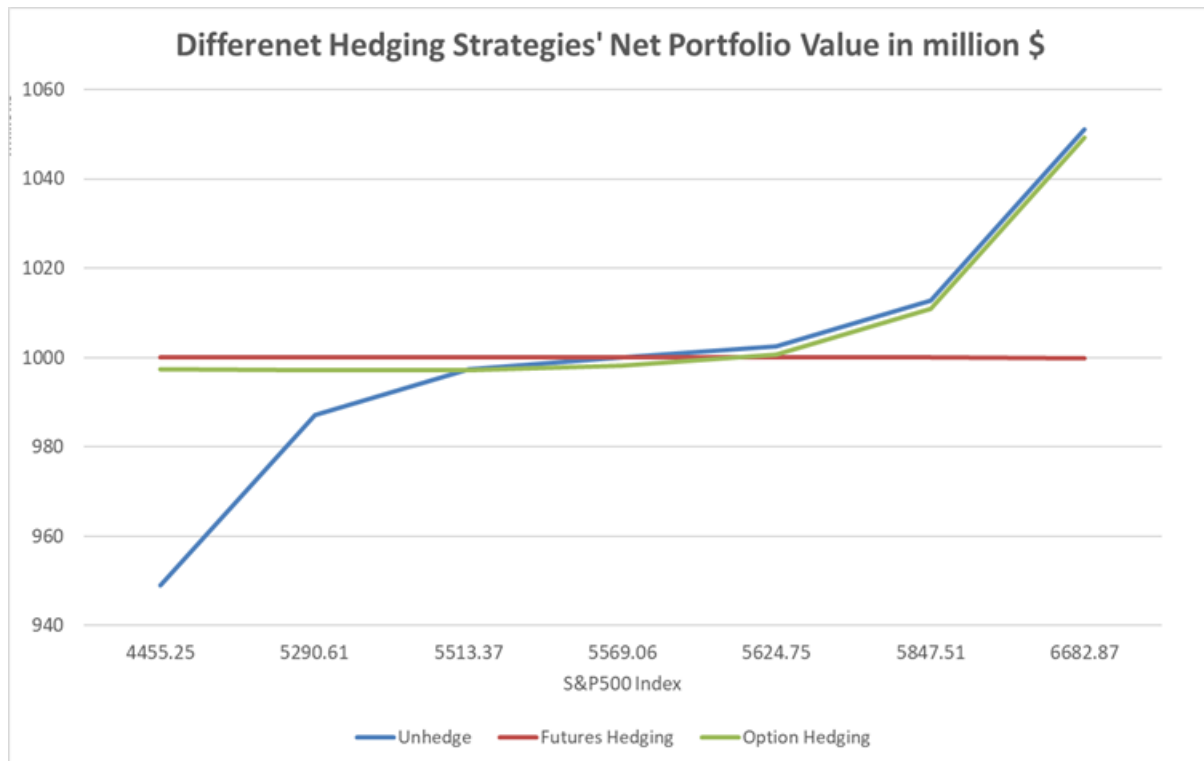
We assume purchasing a slightly out-of-money put option with a 5550 strike price expiring in September. The contract cost \$40.75, and the contract multiplier is \$50. Therefore, the total premium paid is \$40.75 x \$50 x 918 = \$1,870,425.

Hedging Outcome:

S&P500 Change	S&P500 Index	Put Option Value	Total Premium Paid	Net Option Gain/Loss	Unhedged Portfolio Gain/Loss	Hedged Portfolio Gain/Loss	Hedged Portfolio Value
20%	6,682.87	0	-1,870,425	-1,870,425	+51,088,000	+49,217,575	1,049,217,575
5%	5,847.51	0	-1,870,425	-1,870,425	+12,772,000	+10,901,575	1,010,901,575
1%	5,624.75	0	-1,870,425	-1,870,425	+2,554,400	+683,975	1,000,683,975
0%	5,569.06	0	-1,870,425	-1,870,425	0	-1,870,425	998,129,575
-1%	5,513.37	+1,681,317	-1,870,425	-189,108	-2,554,400	-2,743,508	997,256,492
-5%	5,290.61	+11,906,001	-1,870,425	+10,035,576	-12,772,000	-2,736,424	997,263,576
-20%	4,455.25	+50,249,025	-1,870,425	+48,378,600	-51,088,000	-2,709,400	997,290,600

4. Hedging Outcome Comparison

As shown in Graph B.4, the net portfolio value outcomes from different hedging strategies vary a lot. The Unhedge portfolio is risky because it is fully exposed, and its value increases or decreases depending on market performance. In contrast, the net portfolio value remains nearly constant at the initial value when futures hedging is implemented, as this strategy effectively makes the portfolio immune to market movements. Option hedging, on the other hand, sets a maximum loss on the portfolio value but retains the potential to gain from a bullish market.



Graph B.4: Net portfolio value as S&P500 index changes under different hedging strategies.

An unhedged portfolio is the most volatile. The question is which hedging strategy is superior. Comparing the two hedging strategies, futures hedging is a cheaper and safer approach as it eliminates all changes in portfolio value. Conversely, option hedging incurs a higher cost in the form of the option premium, resulting in a lower net portfolio value when the S&P 500 index declines. However, the option grants the holders the right to benefit from favourable market movements. Therefore, when selecting a hedging strategy, the risk manager should consider both the anticipated market movement and the hedging cost. If there is a high possibility that the S&P 500 index is anticipated to drop, a futures hedge may be adequate due to its protection at a lower cost, while the forgone opportunity for gain does not matter much. If the S&P 500 index is anticipated to either drop or rise drastically, option hedging will be a better choice as it limits the loss while still providing the opportunity to benefit. Thus, hedging costs and market forecasts are the critical factors affecting the selection of a risk management strategy.

5. Managing Market Risk in Silo

The Silo management of market risk in investment institutions refers to a decentralized risk management approach, where individual investment departments, trading teams, or fund managers assess, monitor, and manage market risk independently based on the characteristics of their portfolios, risk tolerance, and performance objectives. This often results in a lack of communication and coordination with other departments or the company's central risk management (Przetacznik, Sylwia 2022). Although this management model offers certain flexibility, autonomy, and the ability to respond quickly to the market, it also has significant flaws. For instance, fragmented risk management perspectives, redundant or conflicting

hedging decisions, low cost efficiency, and difficulty in accurately grasping the overall risk exposure. These issues can be further magnified in the actual use of strategies such as futures hedging, options hedging, and unhedged positions.

When departments engage in hedging operations using derivative financial instruments such as futures or options, these issues become more pronounced. Different departments may use similar futures contracts for directional hedging. Still, a lack of communication can lead to redundant hedging or even opposite operations, reducing overall hedging efficiency and increasing cost burdens. Although options strategies provide high flexibility, a lack of unified risk perspective can also result in redundant hedging against volatility or Delta, weakening the effectiveness of the original strategy (*Rebuffed: A Closer Look at Options-Based Strategies*, n.d.). Additionally, some departments may choose not to hedge due to strategic considerations, and if this unhedged position were to occur, it could lead to potential systemic risks. Overall, while a siloed risk management model can enhance the flexibility and responsiveness of individual units, it also results in opacity of overall risk information, duplication of resource allocation, and difficulties in integrated management, which in turn lowers the overall risk control capability and economic benefits of the organisation. Thus, some considerations must be accounted for when managing market risk in silo.

Cost Assessment: The cost structures for various hedging strategies vary because of transaction fees, margin requirements, interest payments, and capital occupation costs (Taylor, 2024). For example, while futures contracts do not require a premium payment, they involve maintenance margins and potential margin calls. On the other hand, options require upfront premiums, and their costs can significantly increase due to changes in volatility. It may lead to resource duplication, overlapping hedging effects, or blind spots if each unit decides on its own hedging method without coordination. Therefore, the company should comprehensively assess the cost-effectiveness of various tools, ensure the rational allocation and efficient usage of funds, and identify the most cost-effective strategies under specific market conditions.

Liquidity Assessment: The market liquidity of hedging instruments directly affects their operability and the flexibility of risk response (*Liquidity Assessment: Ensuring Solvency: The Criticality of Liquidity Assessment in Account Analysis - FasterCapital*, n.d.). If the selected tools lack market depth or have low trading activity, enterprises may encounter slippage, liquidity discounts, or the inability to trade promptly when establishing or closing positions, which will severely weaken the effectiveness of the hedging strategy. Liquidity risk tends to be amplified, especially during periods of significant market volatility and increased hedging demands. Thus, during the decision-making process, businesses should thoroughly evaluate the liquidity indicators related to the tools, such as trading volume, market depth and spreads, ensuring they can adjust positions at critical moments flexibly.

Consideration of Strategy Complexity: Certain derivative financial instruments, such as options portfolios, dynamic hedging, and structured products, possess high design complexity and execution difficulty (Tefi Alonso, 2022). The implementation process requires specialised risk modelling, valuation analysis, real-time monitoring, and trading execution capabilities. If a business or unit lacks the corresponding systemic tools, knowledge system, or professional teams, errors in judgment, model inaccuracies, or operational mistakes may occur during execution, resulting in ineffective risk hedging and potentially amplifying losses. Thus, before selecting complex strategies, an objective assessment should be made regarding whether the

organisation possesses the necessary implementation capabilities. If needed, external consultants or training resources should be introduced, along with small-scale pilot testing to control risks.

Communication and Collaboration Mechanisms: In a unit-based environment, although each investment unit is independent, promoting higher levels of communication and collaboration in risk management will significantly enhance the overall control over market risks (Castañer & Oliveira, 2020). Regular market opinion exchanges, also known as strategy assessment meetings, enable team members to learn from each other's risk response experiences, identify potential cross-risks, and avoid redundant hedging or conflicting strategies by establishing a cross-departmental risk information-sharing mechanism. A good internal collaboration atmosphere not only helps improve hedging efficiency but also contributes to building a more robust and unified corporate risk culture.

Compliance and Regulatory Requirements: The implementation of the hedging strategy must strictly adhere to relevant laws and regulations, as well as compliance guidelines established by regulatory agencies (Kezia Farnham, 2025). Different business units may face varying compliance requirements due to the differences in financial products, regional markets, or types of customers involved. For example, some strategies may be subject to restrictions regarding leverage ratios, capital adequacy ratios, reporting disclosures, or customer protection clauses. If sufficient regulatory review and compliance rehearsals are not conducted before implementation, it may not only lead to regulatory violations and fines but could also cause long-term damage to the company's reputation. Therefore, enterprises should have their compliance departments review the legality of risk strategies before hedging, ensuring that every risk control measure is executed in compliance with the legal framework.

6. Managing Firmwide Risk

Firm-wide market risk refers to the aggregation of all types of risks across the entire organisation. In fact, firmwide risk management is crucial because risks in different business units are often interrelated. Therefore, ignoring these interdependencies can lead to inefficient capital allocation or an underestimation of the overall risk exposure.

Firmwide risk management, also known as integrated risk management, is typically implemented within an economic capital (EC) framework. This framework provides institutions with a method to consistently assess risk, allowing them to allocate capital to cover the financial consequences of risk-taking activities. Hence, when managing firm-wide market risk, several critical factors across the organisation must be taken into account:

Risk Appetite

First, the company should consider its risk appetite, which defines how much risk it is willing to take in pursuit of its strategic objectives and business plan, both quantitatively and qualitatively (Reiner, 2012). In this regard, an effective Risk Appetite Framework (RAF) can help manage firm-wide risk by clearly defining boundaries for market risk-taking. Additionally, the RAF assists in prioritising the allocation of economic capital and liquidity based on market risk exposure. For example, to align with a conservative risk appetite, we can

establish clear limits, such as maintaining a portfolio beta below 0.30 and a VaR_{99%} of less than \$45 million. Consequently, our portfolio complies with these thresholds, as it records an average beta of 0.25544 and a VaR_{99%} of approximately \$40 million. Therefore, this structured approach not only ensures control over overall risk exposure but also enables us to identify areas where more aggressive investments may be pursued, provided they offer higher risk-adjusted returns.

Risk Profile

The company should also consider its risk profile, as it reflects overall risk exposures, financial strength, and the capacity to manage those risks. Specifically, a risk profile provides a quantitative assessment of various risk categories, including strategic, operational, financial, as well as compliance, legal, and regulatory risks (Pratt, 2025). By conducting a risk profile, companies can effectively identify, determine, and prioritise risks, thereby improving budgeting and resource allocation. In our case, despite our portfolio having a low overall beta, its return distribution is negatively skewed with high kurtosis, indicating a "fat left tail". In other words, there is a greater likelihood of extreme losses than extreme gains. Therefore, risk profiling is necessary to identify hidden vulnerabilities. For instance, the Hang Seng Tech Index (HSTECH) in our portfolio, which has the highest beta of 0.62, would be flagged as a concentration risk hotspot. Therefore, this component would require closer monitoring or the implementation of additional hedging measures.

Legal and Regulatory Environment

Companies must comply with various financial regulations, such as Basel III, EMIR, the Dodd-Frank Act, or local Chinese rules, depending on their operating location (Mohanty et al., 2018). These regulations ensure that all parts of a company use appropriate hedging tools, report transactions transparently, and keep risks within acceptable limits. As a result, they play a vital role in enabling consistent and effective risk management across the entire organisation. In our case, since the proposed hedging strategy involves E-mini S&P 500 futures and options, which fall under the jurisdiction of the Dodd-Frank Act, we must ensure full regulatory compliance. Specifically, this includes proper reporting through approved channels, meeting margin requirements, and ensuring that all contracts align with Basel III's capital and liquidity standards. For instance, if multinational firms like HSBC or Alibaba were to manage a similar portfolio, they would need robust cross-border compliance mechanisms to coordinate hedging of Hong Kong-listed assets using U.S.-regulated derivatives. Therefore, legal and regulatory considerations are not only technological needs but also essential to guaranteeing firm-wide uniformity and integrity in risk mitigation procedures.

Market Conditions

When managing firm-wide risks, companies should consider market conditions, including economic factors, inflation, interest rates, and economic growth, as they significantly impact a company's operations and financial performance (Team, 2024). Furthermore, companies that operate worldwide should also consider political events, such as elections, trade policy, and geopolitical concerns, as these events can disrupt financial markets and increase volatility.

Although our portfolio has only moderate exposure to U.S. equity performance, it remains sensitive to broader macroeconomic and geopolitical shifts. For example, rising U.S. interest rates or escalating tensions between China and the U.S. could adversely affect the Hang Seng China Enterprises Index (HSCEI) or the Hang Seng China-Affiliated Corporations Index (HSCCI). Therefore, it is essential to continuously assess whether our current hedging tools, such as S&P 500 futures and options, remain effective, especially if correlations between the indices and global markets shift over time.

Liquidity and Operational Preparedness

To enhance companies' preparedness for hedging activities and manage firm-wide risks more effectively, sufficient liquidity is essential to cover margin and collateral obligations, thereby funding hedging activities (Holyk, 2025). In our case, since our portfolio utilises E-mini S&P 500 futures and options to mitigate market risk, these instruments require daily margin and collateral payments, particularly during periods of extreme volatility. This requires maintaining a buffer of high-quality liquid assets, as introduced by the Basel III framework, to meet future margin calls and other liquidity requirements. For example, if the S&P 500 undergoes a sharp downturn and our futures contracts incur losses, the portfolio might experience immediate margin calls. Moreover, a lack of liquidity could compel the company to postpone collateral or sell assets at disadvantageous prices, putting its operations and reputation at jeopardy. However, by maintaining adequate liquidity buffers, the firm can promptly respond to such demands, protect its hedging positions, and maintain investor confidence. Therefore, liquidity and operational preparedness must be taken into account when managing firm-wide risk.

(3,960 net word count)

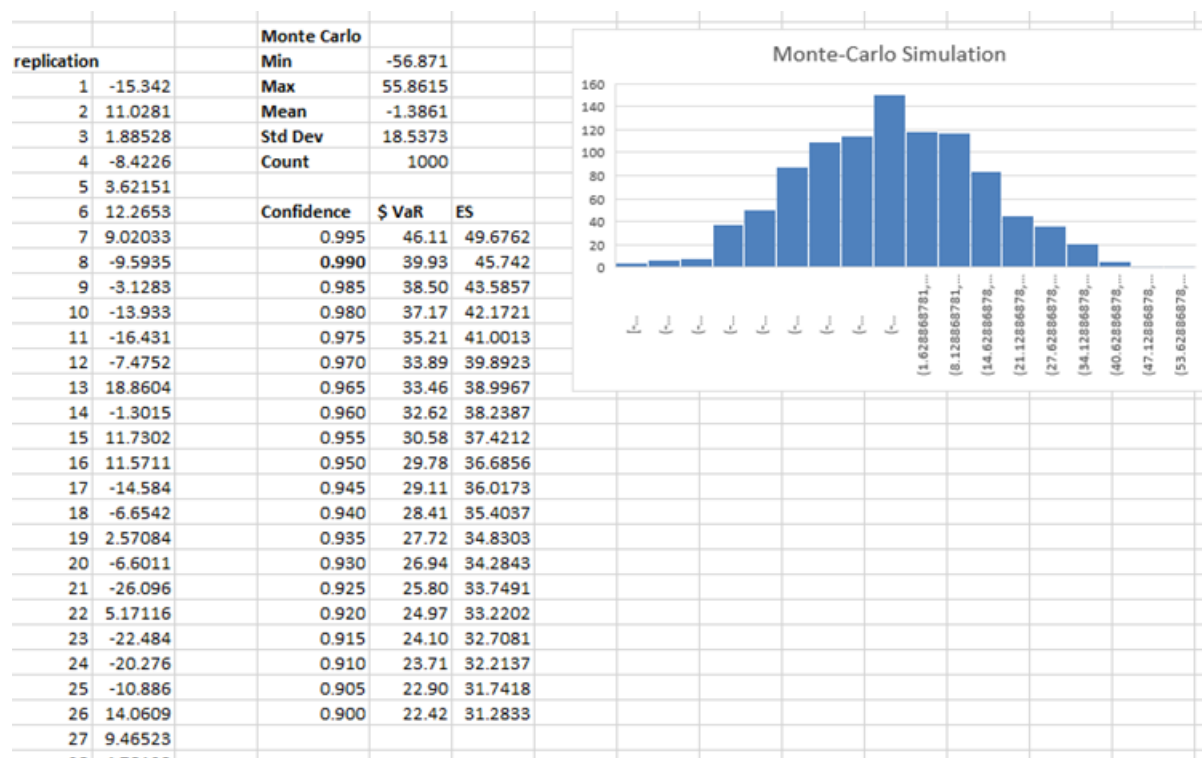
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E. Appendix

Appendix: Monte-Carlo Simulation result



Appendix: Options Hedging

Strike	Open	High	Low	Last	Change	Bid	Ask	Volume	Open Int	Premium	Time	Links
5,300.00P	25.25	25.25	25.25	25.25	-0.25	24.75	25.50	1	8,544	1,262.50	18:53 ET	⌵
5,325.00P	27.25	27.25	25.75	26.75	-0.25	26.00	26.25	150	318	1,287.50	07/01/25	⌵
5,350.00P	28.25	28.25	27.75	27.75	-0.25	27.25	27.75	59	5,189	1,412.50	07/01/25	⌵
5,375.00P	31.75	31.75	29.00	29.00	-0.25	28.50	29.00	56	372	1,475.00	07/01/25	⌵
5,400.00P	31.75	32.75	30.25	30.50	-0.25	29.75	30.50	798	10,266	1,537.50	07/01/25	⌵
5,425.00P	33.50	34.00	32.00	32.00	-0.25	31.25	32.00	42	207	1,625.00	07/01/25	⌵
5,450.00P	35.25	35.25	32.50	33.50	-0.25	32.75	33.50	55	3,486	1,687.50	07/01/25	⌵
5,475.00P	36.75	38.25	35.00	35.00	-0.25	34.50	35.00	162	187	1,912.50	07/01/25	⌵
5,500.00P	37.75	40.25	35.00	36.75	-0.25	36.00	36.75	1,993	19,795	1,800.00	07/01/25	⌵
5,525.00P	39.25	39.25	38.00	38.50	-0.25	37.75	38.50	32	260	1,900.00	07/01/25	⌵
5,550.00P	42.75	42.75	39.25	40.50	-0.25	39.75	40.25	206	4,336	1,962.50	07/01/25	⌵
5,575.00P	45.00	46.25	41.75	42.50	-0.25	41.75	42.25	27	188	2,087.50	07/01/25	⌵
5,600.00P	45.50	48.00	43.25	44.50	-0.25	43.75	44.50	428	8,666	2,237.50	07/01/25	⌵
5,625.00P	49.50	49.50	46.75	46.75	-0.25	46.00	46.50	195	304	2,450.00	07/01/25	⌵
5,650.00P	53.25	53.25	47.50	49.00	-0.25	48.25	49.00	472	5,171	2,462.50	07/01/25	⌵
5,675.00P	53.75	54.00	50.25	51.50	unch	50.50	51.25	23	1,239	2,512.50	07/01/25	⌵
5,700.00P	54.00	57.25	51.50	54.25	+0.25	53.00	53.75	1,735	8,928	2,625.00	07/01/25	⌵
5,720.00P	60.50	60.50	55.50	56.50	+0.25	55.25	56.00	27	624	2,775.00	07/01/25	⌵
5,725.00P	60.00	61.50	55.75	57.00	+0.25	55.75	56.50	53	467	2,787.50	07/01/25	⌵
5,750.00P	62.75	63.25	58.00	59.75	+0.25	58.50	59.25	2,521	4,670	2,900.00	07/01/25	⌵

Note: It was \$40.75 on 1/7/2025.